

Measurement as a Move

A quantum chess variant with bounded branching, adversarial collapse,
and a minimal observation language

L. G. Meredith
F1R3FLY.io

September 2026

Abstract

We report the design of a quantum variant of chess in which a player may play a superposition of at most m classical moves, and may additionally elect, as a whole turn, to force a measurement of the board. The literature contains two mature lineages — superposition of *moves* in the possibilistic style of Dorbec and Mhalla, and superposition of *positions* under unitary dynamics in the style of Cantwell — but in every design we have found, measurement is *triggered* by a rule rather than *elected* by a player. The elective measurement move is the novel element, and it forces a series of further choices which this note records: who resolves the measurement, what may be asked, and what the branching bound counts. The design we arrive at resolves the measurement adversarially (the opponent supplies the answer), which keeps the game free of chance and answers the recent result of Scholten, Westerbaan and Samardjiska that quantum moves confer no advantage against perfect classical play. It equips the measurement move with a deliberately small observation language whose atoms are patterns against the board and whose two connectives are restricted so that one merges and one correlates, neither able to impersonate the other. We give the grammar, the semantics, an optional phase layer under which interference has the unusually legible reading of *menu deletion*, a sketch of the player as an f1r3lang process, and a list of open questions, chief among them whether the branching bound is a capability at all — that is, whether the optimal m for a perfect player exceeds one. Three appendices work a nine-ply trace in the Italian Game in which both players superpose both sides of a book fork, give a ZX-calculus semantics for traces under which a trace compiles to one interferometer per Q-move that closes, and verify the two against each other.

Contents

1	Introduction	2
2	Prior art	3
2.1	Superposition of moves: the possibilistic lineage	3
2.2	Superposition of positions: the unitary lineage	3
2.3	Other variants	4
2.4	The quantum Zermelo result	4
2.5	The gap	4
3	The design axes	4

4	The design	5
4.1	Sets, with amplitudes as an optional layer	5
4.2	The opponent resolves	6
4.3	Measurement is a whole turn	6
4.4	Bounded branching	6
4.5	No check	7
5	The observation language	7
5.1	A correction worth stating plainly	7
5.2	Atoms	7
5.3	Indices and wildcards: a lattice of observables	8
5.4	Two connectives, restricted so neither can impersonate the other	8
5.5	The grammar	9
5.6	Safety is semantic, not syntactic	9
6	The move language	10
6.1	Why a move needs a source annotation	10
6.2	Signs on branches	11
6.3	Legality does not live in the grammar	11
7	Semantics	11
8	Phase: an optional layer	12
9	Players as processes	13
9.1	Typing, and the player whose source we do not have	14
10	The branching bound as a ranking	14
11	Open questions	15
12	Conclusion	16
A	A worked trace: the Italian transposition diamond	17
A.1	Choosing an opening	17
A.2	Notation	17
A.3	The trace	17
A.4	Evolution of the realisation set	18
A.5	Adding signs	18
A.6	The measurement variant	19
B	A ZX semantics for game traces	19
B.1	The question	19
B.2	Conventions	19
B.3	Registers	19
B.4	The translation	19
B.5	Unitarity and coherence	20
B.6	The trace translated	21
B.7	The numbers	22
B.8	The measurement variant	22
B.9	The hardware reading	22
B.10	What this does and does not show	23

1 Introduction

The proposal under study is easy to state. A player may, on their turn, play a *superposition* of classical chess moves rather than a single one, subject to a bound m on how many moves may be superposed. A player may alternatively spend their turn forcing a *measurement*, collapsing some of the resulting indeterminacy.

The first half of that proposal is well covered by existing work, twice over and from two different directions. The second half appears to be unclaimed, and it is where the design interest lies. In every quantum board game we have located, measurement happens because a rule says it must: because a cycle of entanglement has closed, because a capture has been attempted, because a stone has been placed adjacent to a superposed stone. Nowhere is measurement a move a player may choose to make, at a tempo cost, naming what is to be asked.

Making measurement elective turns out not to be a local addition. It interacts with almost every other design choice, and pursuing those interactions is what produced the design reported here. Section 2 reviews the prior art and locates the gap precisely. Section 3 lays out the design space as a set of independent axes. Section 4 records the choices we made and why. Sections 5 and 6 give the observation and move languages, which are the two artefacts of the design most likely to go wrong by accident and are therefore treated in detail. Section 7 gives the semantics. Section 8 describes an optional phase layer. Section 9 sketches the player as a process in `f1r3lang`, connecting this variant to the multi-agent chess player already under development. Section 10 describes the branching bound as a tournament structure. Section 11 collects the open questions.

A word on what kind of document this is. The design is not finished. Several choices below are recorded as decided because they were decided in the course of the dialogue that produced this note, and several others are recorded as open because they turn on facts about the resulting game that only play or search will supply. We have tried to be explicit about which is which.

2 Prior art

2.1 Superposition of moves: the possibilistic lineage

Dorbec and Mhalla’s *Toward Quantum Combinatorial Games* [7] is the general form of the first half of our proposal. They generalise Goff’s quantum tic-tac-toe [8] to arbitrary combinatorial games by allowing players to play superpositions of moves. In their terminology a *Q-move* is a set of classical moves; a *run* is a choice of one classical move from each Q-move played so far; and if a run forms a legal sequence in the underlying classical game, the position it reaches is a *realisation*. They propose several rulesets, differing in when superposed and unsuperposed moves may be played, and prove that the rulesets separate: the same classical game can have different outcomes under different rulesets. Their worked case is Nim.

Crucially, this lineage is *possibilistic*. A superposition is a set of realisations, carrying no amplitudes. There is no Born rule and no interference. What quantum mechanics contributes is the branching structure and the collapse discipline, not the linear algebra.

Burke, Ferland and Teng [2, 4] take up the framework, concentrating on what they call flavour D, in which classical moves are always available as one-wide superpositions, so that the quantum game contains the classical one. Their results are about complexity: starting

from superpositions, Nim becomes Σ_2^P -hard and Undirected Geography becomes PSPACE-complete, and they exhibit abstract PSPACE-complete games whose quantum extensions fall to each level of the polynomial hierarchy. They also make a methodological observation that matters for us: in quantum combinatorial games the *description* of a move is part of the game definition, because move description affects collapse. Two moves with the same effect on a classical position may filter a superposition differently.

The same group’s *Transverse Wave* [3] and the recent *Cheqqers* [6], a quantum checkers built in flavour D, extend the lineage to further games.

2.2 Superposition of positions: the unitary lineage

Cantwell’s Quantum Chess [5] is the serious engineering of the other approach. The board is a quantum register, one qubit per square, and moves are unitaries: split moves send a piece to two targets, merge moves recombine, and the resulting dynamics exhibit superposition, entanglement and interference. Measurement is present but is *forced by move type* — the excluded and capture variants of each move require a measurement so that the engine can unambiguously assign a piece to the target square. A measurement rule additionally serves to keep the superposition small enough for the game to remain tractable classically and comprehensible to a player.

Cantwell’s treatment of the win condition is the one piece of his design we adopt wholesale. There is no check; a player wins by capturing the opponent’s king with certainty, and while both kings have nonzero probability of being on the board the game continues. Any design that admits a possibilistic notion of check — mate only if mated in every realisation — makes superposition a free escape from every mating net, and the game collapses.

2.3 Other variants

Akl’s Quantum Chess [1] superposes piece *identity* rather than position: a piece is a superposition of types, and collapses to one of them when touched. Goff’s quantum tic-tac-toe [8] forces collapse when a cycle of entanglement closes, and — a detail we return to — the player who did *not* create the cycle chooses where the collapse begins. Quantum Go exists in several forms [9], with collapse triggered by adjacency.

2.4 The quantum Zermelo result

Scholten, Westerbaan and Samardjiska [10] give the result any new design in this space must answer. Instantiating Zermelo’s theorem in the quantum setting, they show that quantum effects convey no advantage against a classical player who plays a perfect classical strategy; the quantum player’s edge arises only from amplifying the classical player’s mistakes.

The lesson we take is not that quantum variants are pointless, but that a design which adds *randomness* adds nothing a perfect opponent must respect. If the quantum machinery is to matter, it must change the option structure of the game rather than the distribution over outcomes.

2.5 The gap

Collecting the above: superposition of moves is done, superposition of positions is done, collapse disciplines of at least four kinds are done, and the complexity landscape is partly mapped. What is not done, as far as we can determine, is any of the following.

- **Elective measurement.** No design lets a player spend a turn to measure. Measurement is always a consequence of some other action.

- **A chosen observable.** Where measurement occurs, what is measured is fixed by the rule that triggered it. No design gives the player a language in which to say what should be asked.
- **The branching bound as a first-class parameter.** Cantwell bounds superposition indirectly, via the measurement rule. Nobody treats the width of a superposition as a declared resource, still less as a difficulty setting.

These three are the content of the present design, and they are not independent: giving the player a measurement move forces the question of what may be asked, and the answer to that question is what makes the bound tractable.

3 The design axes

Before recording choices we lay out the space. The axes below are close to independent, and a design is a point in their product.

Axis 1: amplitudes or sets. The fork that conditions everything else. Sets give a combinatorial game: no chance, Zermelo applies, outcome classes are well defined, and a measurement means selecting among realisations. Amplitudes buy interference, at the price of having to make each branching move unitary — which is exactly where branch-dependent legality causes trouble, since a move legal in one realisation and illegal in another is not obviously a unitary at all.

Axis 2: who resolves a measurement. Three candidates: nature, by the Born rule; the measuring player; or the opponent. The third is the one that preserves the character of chess, since it introduces no chance and no hidden information, and it prices the measurement move automatically — forcing a measurement means handing the opponent a menu.

Axis 3: what may be measured. The whole board; a single square’s occupancy; a single piece’s position; or a *predicate* over the board. The last is the general case and contains the others. Its importance is not expressive: it is that the *coarseness* of the observable is what limits the size of the choice handed over under Axis 2.

Axis 4: what the branching bound counts. Per-move width; total support size, with a forced collapse on overflow; a per-game budget of quantum moves; or a decaying coherence budget. These give materially different games. Per-move width is the one a player can reason about at the board.

Axis 5: whether superposed moves share a source. Cantwell’s split move sends one piece to two targets. Dorbec and Mhalla allow arbitrary sets of moves, so branches may differ in which piece moved, and hence in material and tempo. The latter is more expressive and considerably harder to visualise; most of the bookkeeping pain in the design — castling rights, promotion, captures that occurred in only some branches — originates here.

Axis 6: check and mate. Possibilistic check is fatal, as noted above. Cantwell’s abolition of check in favour of certain king capture is, as far as we can see, the only option that survives contact with superposition.

4 The design

We record the choices, with the reasoning that produced them.

4.1 Sets, with amplitudes as an optional layer

We take the possibilistic semantics as the base design: a position is a set of realisations. This is flavour D in the Burke–Ferland–Teng sense, so classical chess embeds as the $m = 1$ fragment and a player may always decline to branch.

The reason is not conservatism but a consequence of Axis 2, developed in §4.2: under adversarial resolution, amplitudes carry no strategic information. Only the support matters. So the set semantics is not an approximation to the amplitude semantics; it is the whole of what matters, unless phase is added deliberately so that cancellation can empty a cell of the opponent’s menu. That is the optional layer of §8.

4.2 The opponent resolves

A measurement’s outcome is chosen by the player who did not call for it. Only an answer with nonzero support is legal.

Three consequences.

First, the game contains no chance. It remains a combinatorial game with perfect information, Zermelo applies, and outcome classes are well defined.

Second, the Scholten–Westerbaan–Samardjiska result is answered rather than dodged. There is no statistical edge to be had, because there are no statistics; whatever advantage the quantum machinery confers must show up in the structure of available options.

Third, the criterion for calling a measurement is a minimum over the outcomes: one measures only when *every* legal answer is acceptable. This is computable at the board, teaches itself, and makes the measurement move rare and decisive rather than routine.

Remark 4.1 (The entanglement objection, and its resolution). The natural objection to adversarial resolution is that the board state is globally entangled, so that handing the opponent a collapse hands them the game. In the set semantics the objection has real force: a set of realisations is a subset of a product, and a subset is generically not a product, so correlation is the default rather than something the branches must work up to.

The resolution is that the size of the menu handed over is set by the observable’s spectrum, not by the entanglement. A measurement that projects onto a rank-one subspace — a whole realisation — does hand over the game. A measurement that projects onto a large eigenspace hands over one bit, restricting the state to the realisations consistent with the answer while leaving it superposed and entangled. The *effect* of the measurement is global, because correlations carry it everywhere. The *choice* handed over is one bit.

Hence the observation language must be coarse by construction, and §5 is written to make it so. Full-board collapse is not in the language.

4.3 Measurement is a whole turn

The turn structure is:

1. The player to move declares **measure** φ for a formula φ of the observation language.
2. The opponent answers *yes* or *no*. Only answers with nonzero support are legal.
3. The realisation set is filtered to those realisations consistent with the answer. Correlations propagate the effect across the board.

4. The turn ends. The opponent now plays an ordinary turn.

The asymmetry between steps 1 and 2 is the design’s load-bearing detail. The measuring player chooses *what is asked*; the opponent chooses *the answer*. Choosing the observable is where skill enters; choosing the answer is the price. If the opponent chose the question too, one would never measure; if the measuring player chose the answer, one would do nothing else.

The opponent’s answer is a response *inside* the measuring player’s turn, not a turn of their own, so the tempo cost of measuring is exactly one ply.

4.4 Bounded branching

A Q-move superposes at most m distinct classical moves. We take m as a declared parameter of the match rather than a constant of the game; see §10.

4.5 No check

Following Cantwell: no check, no checkmate, no movement restrictions that depend on them. A player wins by capturing the opponent’s king with certainty — in the set semantics, in every surviving realisation.

5 The observation language

By Remark 4.1 the observation language is where the design is kept safe, so we develop it carefully. The design goal is unusual: we want the *smallest* language that maximises the interference available, subject to never naming a rank-one projector on a typical board.

5.1 A correction worth stating plainly

Every predicate built from occupancy atoms is diagonal in the configuration basis. A diagonal measurement cannot *create* interference; it can only destroy it. So the language’s job is not to generate interference but to govern how much coherence survives a measurement.

A projector preserves every coherence *within* an eigenspace and destroys every coherence *across* eigenspaces. That reduces the design question to a single one: how large, and how well placed, are the eigenspaces the language can name?

Principle 5.1 (Precision cuts, coarseness preserves). A sharper question separates more branches and destroys more coherence; a coarser question merges more branches and preserves more. Every feature of the observation language should be understood as a position on this dial.

5.2 Atoms

An atom asks for the *value at a square*:

$$atom ::= (i, j)!(piece\ Or\ Empty).$$

The notation is deliberate: this is a pattern against the board agent of the multi-agent chess design, where the board is $\prod_{1 \leq i, j \leq 8} @(i, j)!(\dots)$ with the payload either 0 for an unoccupied square or $*x$ for the channel minted for the piece at that square. The observation language is therefore a fragment of a spatial logic over the board process, and a measurement is the extension of a spatial formula used as a projector. Consequences for implementation are discussed in §9.

Two features of this choice matter.

***empty* is a value, not a negation.** Were emptiness expressed as the negation of occupancy, “this square is empty” would require a disjunction over all thirty-two pieces, which no small language can express. As a value it is an atom.

The value is a colour-and-type, not a piece identity. This is where coherence comes from for free. If one branch has one black rook on a square and another branch has the other black rook there, both satisfy the same atom, and the projector keeps them in one eigenspace with their relation intact. The state must still track identity, for castling rights and for promotion; the *question* cannot see it. That gap between what the state distinguishes and what the language can ask is precisely where coherence survives, and it is indistinguishability doing the work rather than any syntactic device.

5.3 Indices and wildcards: a lattice of observables

Identity is not simply hidden: it is *graded*. Each piece type carries an index, and the index may be a wildcard.

```

color      ::= B | W
type       ::= Pawn( _ | 1..8 )
              | Bishop( _ | L | R )
              | Knight( _ | L | R )
              | Rook( _ | L | R )
              | Queen
              | King
pieceOrEmpty ::= empty | ( color, type ) | ( color, _ )

```

The indices name pieces present at setup and are positional, not identity-tracking: **Pawn(3)** means the pawn that began on the third file, **Rook(L)** the queenside rook, and so on. **Bishop(L|R)** is the happiest case, since light-square and dark-square is a genuine invariant rather than a label. King and Queen are unindexed, there being one of each at setup.

This yields a lattice of observables over any square, ordered by coarseness: **Pawn(3)** is sharpest; **Pawn(_)** merges the eight pawns; **(B, _)** merges every black piece; and the coarsest occupancy question is **(B, _) v (W, _)**. Choosing a question is choosing a point on this lattice, which is to say: the wildcard *is* the dial of Principle 5.1. This is better than a menu of connectives, because it is a single mechanism a player can learn once.

Remark 5.2 (Promotion blurs identity, by construction). Since Queen is unindexed, a promoted queen is indistinguishable from the original. The rule generalises to under-promotion: indices name pieces present at setup, and promoted pieces carry **_** as their standing index. A promoted knight is **Knight(_)** and nothing sharper.

The wildcard therefore does two jobs — wildcard in a query, and standing index of a promoted piece — and the two agree, since **Knight(_)** matches the promoted knight along with the originals and no query singles it out. That coincidence is load-bearing rather than decorative: it is what keeps promotion from requiring its own clause.

It also has a pleasant reading in the game. Promotion converts a sharply nameable piece into a permanently coarse one, so every promotion enlarges the eigenspaces available to later measurements. A pawn that queens buys material and buys coherence.

5.4 Two connectives, restricted so neither can impersonate the other

Negation is unnecessary: the opponent answers yes or no, so asking φ and hearing *no* already projects onto $\neg\varphi$. The complement eigenspace is always reachable, and a negation connective would add nothing.

Of the remaining candidates, the jobs separate cleanly.

- An **atom** is a cut. It is the sharpest instrument available and destroys coherence between the branches it separates.
- **Disjunction** is a merge. Its yes-eigenspace is the span of both disjuncts, so branches differing only in which disjunct holds survive together. This is which-path erasure, and it is the smallest formula that has any. In the slogan of the graded where-clause work: interference is a false disjunction with true disjuncts.
- **Conjunction** is strictly more expressive and its added power is all cutting, since intersecting eigenspaces makes them smaller. It buys reach at the direct expense of what we are trying to maximise, and is excluded.
- The **biconditional** is a correlation probe. Its yes-space contains both the both-true and the both-false configurations, so the answer reveals a *relation* while leaving each atom maximally undetermined. This is the only cheap connective that addresses entanglement without collapsing either party to it.

Left unrestricted, however, each of the two chosen connectives can do the other's job badly. A biconditional between two atoms on the *same* square is true exactly when both are false, since the atoms are mutually exclusive: that is a conjunction of negations in disguise, and it cuts. A disjunction between atoms on *distinct* squares has a yes-space containing configurations where both hold, and is therefore a coarse cut rather than a clean merge.

Hence the restrictions:

```
formula ::= atom
         | atom v atom      -- same square
         | atom <=> atom     -- distinct squares
```

One connective for erasure, one for correlation, neither able to impersonate the other.

Proposition 5.3 (Same-square disjunction is exclusive by construction). *Over a single square the disjuncts are mutually exclusive, so no configuration satisfies both. The yes-space is a clean span of two alternatives, and the merge is as coherence-preserving as a two-outcome measurement can be.*

Thus the restriction imposed to stop disjunction from cutting also made it the sharpest available erasure operator.

Remark 5.4 (Disjunction is not subsumed by wildcards). Wildcards subsume disjunction *within* a type: $(d5)!(B, \text{Rook}(L)) \vee (d5)!(B, \text{Rook}(R))$ is exactly $(d5)!(B, \text{Rook}(_))$. They do not reach across types: $(d5)!(B, \text{Knight}(_)) \vee (d5)!(B, \text{Bishop}(_))$ has no wildcard spelling, since $(B, _)$ is strictly coarser. So the connective's remaining job is cross-type merging on one square, which is narrow enough to describe in a sentence and is the reason it survives.

5.5 The grammar

```
color      ::= B | W
type       ::= Pawn( _ | 1..8 )
           | Bishop( _ | L | R )
           | Knight( _ | L | R )
           | Rook( _ | L | R )
```

	Queen	
	King	
pieceOrEmpty	::= empty (color, type) (color, _)	
square	::= (i, j)	-- 1 <= i,j <= 8
atom	::= square !(pieceOrEmpty)	
formula	::= atom	
	atom v atom	-- same square
	atom <=> atom	-- distinct squares

5.6 Safety is semantic, not syntactic

It is tempting to read the restrictions above as the guarantee that no measurement gives the game away. They are not, and it is worth being explicit about why.

Whether a projector isolates a single realisation is a property of the *current support*, not of the formula. A single atom can be rank-one on today's board; a much larger formula can be rank-eight. Syntax is the wrong instrument for this obligation.

Since the support is capped anyway — that is what the branching bound is for — evaluating a formula against every realisation is cheap. So the safety condition belongs where it can be stated exactly:

Definition 5.5 (Legality of a measurement). *measure* φ is legal in a position S iff

- (i) both answers are live: $|\{s \in S : s \models \varphi\}| \geq 2$ and $|\{s \in S : s \not\models \varphi\}| \geq 2$; and
- (ii) the yes-space contains a coherent pair: at least two realisations satisfying φ that differ only in a way the formula cannot see.

Condition (i) subsumes the prohibition on *vacuous* measurement — a question every realisation already answers the same way, which would burn a ply for nothing — and it also prevents the binary-search degeneracy. Condition (ii) is what makes a measurement an instrument rather than merely a cut, and it may be relaxed if play shows it too restrictive.

We note that tempo already bounds the abuse. With the support capped at, say, eight or sixteen, unrestricted binary search would need three or four plies to collapse the board, which in chess is a catastrophic price. The language restriction is therefore buying legibility and coherence preservation rather than safety; Definition 5.5 buys safety.

6 The move language

6.1 Why a move needs a source annotation

Under superposition, a move described only by source and target square is not well defined, for two reasons.

First, legality is type-dependent: **d1->d5** is legal for a queen or rook and illegal for a knight, and the source square may hold different types in different realisations. The Dorbec–Mhalla machinery requires feasibility to be decidable per realisation in order to know which realisations survive.

Second — and this is the Burke–Ferland–Teng observation — move *description* affects collapse. Two moves with identical source and target but different descriptions filter the realisation set differently, so the description is part of the game and not an artefact of notation.

Hence a classical move carries a pattern at its source:

<code>cmove ::= square !(color, type) -> square [!(color, type)]</code>
--

The move is feasible exactly in those realisations where the source pattern matches. The optional target annotation is required only for under-promotion; a bare target means the piece arrives unchanged.

Remark 6.1 (Wildcards belong here too). The source pattern admits wildcards, on the same grounds as the observation language. `d1!(B,Rook(_)) -> d5` is a move any black rook on d1 may make, feasible across branches that disagree about which rook it is, and therefore preserving coherence between them. Barring wildcards would force every move to be maximally sharp and would decohere the board on every turn.

Remark 6.2 (The source pattern is an atom). Comparing with §5, the source pattern of a `cmove` and the atom of a formula have the same shape. The engine therefore needs one matcher against the board process, used in two roles — as a projector in a measurement and as a pattern in a move.

6.2 Signs on branches

The optional phase layer of §8 requires only that a branch may carry a sign.

```
sign    ::= + | -                -- + may be elided
branch  ::= sign cmove
qmove   ::= { branch , ... , branch } -- width <= m, branches distinct
turn    ::= qmove
          | measure formula
```

Only relative signs matter, so a width-two Q-move carries one meaningful bit; normalising the first branch to + removes the redundant spelling at some cost in symmetry.

Whether the branches of a Q-move must share a source square is Axis 5 and is left open. As written they need not, which is the Dorbec–Mhalla reading. The tighter form

```
qmove ::= square !( color, type ) -> { sign square , ... }
```

is Cantwell’s split move, is far easier to visualise, and eliminates most of the bookkeeping around branches that differ in material.

6.3 Legality does not live in the grammar

The grammar fixes the *shape* of a move: what it claims about its source, where it goes, what arrives. It does not and cannot fix legality, because piece-movement legality is a property of the board — path occupancy, castling rights, en passant windows — and a grammar that expressed only legal moves would have to be indexed by the position, which is not a grammar. Under superposition the point is sharper still: feasibility is not boolean but per-realisation, and that filtering is precisely the mechanism that makes the game quantum.

So `d1!(B,Pawn(3)) -> h8` parses and is refused at the board, which is the correct place to refuse it. The source annotation is not there to make the move legal; it is there to make the move *determinate across branches*.

7 Semantics

We give the set semantics. The amplitude layer is §8.

Definition 7.1 (Position). A *classical position* is a board configuration together with the usual auxiliary state: side to move, castling rights, en passant window, and the counters governing repetition and the fifty-move rule. A *position* is a nonempty finite set S of classical positions, all with the same side to move.

That the side to move is common to all realisations is worth noting: turns are never superposed, so ply parity is classical throughout and the game remains strictly alternating.

Definition 7.2 (Feasibility). A classical move c is *feasible* at $s \in S$ if the source pattern of c matches at s and the resulting move is legal in s by the ordinary rules of chess, as amended by §4 (no check, king capture permitted). Write $\Gamma(s, c)$ for the classical position resulting.

Definition 7.3 (Q-move). Let $Q = \{c_1, \dots, c_k\}$ with $k \leq m$. Q is legal at S if every c_j is feasible at some $s \in S$. The resulting position is

$$S' = \{\Gamma(s, c_j) : s \in S, 1 \leq j \leq k, c_j \text{ feasible at } s\}.$$

Infeasible pairings simply do not contribute, which is the collapse-on- infeasibility of the Dorbec–Mhalla framework. Note that $|S'|$ may exceed $m \cdot |S|$ in no case, and is often much smaller, since distinct pairings frequently reconverge on the same classical position.

Definition 7.4 (Measurement). Let φ be a formula legal at S per Definition 5.5. The opponent selects $b \in \{\top, \perp\}$, and the resulting position is

$$S' = \{s \in S : s \models \varphi \text{ iff } b = \top\}.$$

Measurement is filtering, not collapse to a point. The conditional state remains a superposition, generally still entangled, and the engine must maintain it rather than reduce to a single board. This is the representational cost of coarse observables, and it means a measurement does not necessarily shrink the support very much — which is exactly why the branching bound must do its own work rather than relying on measurements to keep the state small.

Definition 7.5 (Win condition). A player wins when the opponent’s king is absent from every $s \in S$. There is no check and no checkmate.

Remark 7.6 (Overflow). If $|S'|$ exceeds the support cap, some forced reduction is required. The options are a forced measurement chosen by the player to move, a forced measurement chosen by the opponent, or a rule declaring the Q-move illegal. The third is cleanest and keeps the bound a property the player must plan around rather than a trap. We have not settled this; see Question 6.

8 Phase: an optional layer

Adding amplitudes to a game whose measurements are resolved adversarially has an unusual and, we think, attractive consequence.

Under Born resolution, phase shifts probability around, and a player must reason about distributions. This is where every quantum variant loses its human audience.

Under adversarial resolution, amplitudes are invisible to strategy except in one way: when two branches cancel to zero, that configuration leaves the support, and the opponent can no longer choose it.

Principle 8.1 (Interference is menu deletion). With the opponent resolving, the sole strategic effect of relative phase is the removal of options from the player about to choose. A player never computes a probability; they ask whether a cell of the opponent’s menu can be emptied.

This is a chess-shaped question, which is the point.

The mechanism. A Q-move declares a sign on a branch. Nothing else changes syntactically. For the sign to matter, two distinct move sequences must reach the same configuration, so that their amplitudes land in the same cell and may sum to zero. Chess supplies such reconvergence naturally, through transpositions and through split moves that later remerge on a square.

The obstruction, which is familiar. The surviving rest of the board is a which-path record: two branches recombine only if *everything else* about them agrees. On a chess position that is a stringent condition, and it is what keeps the mechanic rare rather than dominant. This is the same phenomenon identified in the graded where-clause setting, where the residual message left by a race is precisely the record that prevents recombination, and where recombination is recovered only when the racing alternatives are indistinguishable.

An unusual simplification. Because nothing is ever sampled, norms do not matter. Amplitudes need only be tracked up to whether they vanish, so a ring with additive inverses suffices; signs alone give ± 1 and cancellation with integer bookkeeping. There is no unitarity discipline to enforce, no normalisation, and no post-selection hazard. The entire apparatus that makes a quantum interpretation of a process calculus difficult is unnecessary here, precisely because no distribution is ever read out.

A risk. A bare sign may be too weak. With ± 1 on integer path counts, cancellation requires two branches to arrive with opposite signs and equal magnitude, which may be a coincidence a player cannot reliably engineer. If so, the escalation is to let the sign be chosen at *merge* time rather than at split time, or to let a measurement's yes-answer carry a sign. Both are larger changes, and we would start with the sign-at-split version and escalate only if the mechanic proves inert. See Question 3.

9 Players as processes

This variant is intended to be playable within the multi-agent chess design, in which each piece of a colour is an agent and the board is itself an agent holding a deposit per square.

A player picking up a piece is naturally written as a sum of receives:

$$\text{Player}(\text{color}) = \sum_{\text{type}} \sum_{\text{pos}} \text{for}((\text{color}, \text{type}) \leftarrow \text{pos}) \{ \text{movePieceAccordingToStrategy}(\dots) \}$$

with the receives linear, as elsewhere in this programme.

Three observations.

The receive is the atom in a third role. $\text{for}((\text{color}, \text{type}) \leftarrow \text{pos})$ is a pattern match against the board deposit $@(i, j)!(\dots)$ — the same shape as the observation atom and the move's source pattern. So one construct appears three times: as a projector in a measurement, as a pattern in a move, and as a binder in a pickup. Wildcards carry over in all three roles, and a pickup with a wildcard grasps a piece without resolving which one, preserving coherence between those branches.

The sum is a race, and the Q-move is that race with a bound. A player offering many receives against a contended board is the general race — bags of consumes against deposits, with admissible pairings. A Q-move is that race with a width bound and, optionally, signs on the branches. The sign therefore belongs on the binder rather than on a separate move

term, which is the same conclusion the graded where-clause work reached by a different route: the control a programmer needs in order to arrange interference belongs where the branching happens.

Measuring and non-measuring interaction must remain distinct. A measuring receive and a non-measuring one differ in what they do to the *rest* of the board, not merely to the square they touch: the former filters the realisation set, the latter leaves it intact across the branches the pattern covers. Same pattern, different effect on state the pattern never mentions. In this game the separation is forced anyway, since measurement consumes a whole turn and invites an answer from the opponent, so the turn structures differ. A general account of measuring versus non-measuring interaction — plausibly a polymorphic interaction with a measure primitive — is a broader question for interactive GSLTs and is out of scope here.

9.1 Typing, and the player whose source we do not have

One may ask for a typing discipline under which a player process is well-typed iff it never makes an illegal move. This is attractive: a well-typed bot needs no rules oracle, which matters when sixteen piece-agents are proposing moves concurrently and one would rather not round-trip each proposal through a referee.

Two caveats.

First, the discipline stratifies. Piece-movement geometry is local to the bound pattern and the target, so a static type can plausibly carry it. Path occupancy, castling rights and their kin are properties of the whole board, so a type ruling them out must be indexed by board state. Under superposition the position is a set of realisations, and the discipline must then choose: legal in *every* realisation is too strong and makes most Q-moves untypable; legal in *some* is the feasibility condition we want, but is not a static property. The resulting stratification — one fragment decidable by inspection, one not — mirrors the stratification found in the graded where-clause work between the modality-free fragment and the rest.

Second, and decisively: a human player is not a term. Well-typedness is a property of a process, and there is no source to inspect when the player is a person. So typing cannot be the game’s legality story.

The resolution is to make the referee’s check the *definition* and the type system a static approximation of it: a player process is well-typed iff every move it can emit would pass the referee. Then legality is defined uniformly over humans and bots, the typing becomes a soundness theorem about bots rather than a rule of the game, and there is no risk of a typed bot being legal by a standard the referee does not share.

For a human, the corresponding obligation is discharged by having read the rules and consented to play, and the check happens at the referee when a move arrives. The referee’s rejections are thus the human-facing half of the type error and should read that way. Definition 5.5 in particular has no counterpart in chess, and a player will encounter it without knowing why; the message should say so.

10 The branching bound as a ranking

The bound m is a parameter of the match, not a constant of the game, and this suggests a tournament structure: a player who succeeds repeatedly at bound m advances to play at bound $m + 1$.

The idea is that the rating is a bound on the state space rather than a number attached to a history. Elo measures how well one plays; m measures how much superposition one can

hold. These are different quantities and the second is the more legible.

The ladder teaches. At $m = 1$ the game is chess. At $m = 2$ it is chess plus the smallest possible quantum move, and the measurement move is immediately comprehensible because the menu has at most two cells. Each rung adds one branch to hold. The difficulty gradient is therefore a genuine cognitive quantity rather than an opponent-strength proxy, which is rare in game design.

Bots and humans become comparable in a meaningful way. A bot's rung says something real about its search: holding m branches under adversarial resolution means evaluating a minimum over a menu at each measurement, which is a different computation from deepening a classical search. A bot that wins only by raising m is not better at chess; it is better at bookkeeping. Worth separating explicitly.

The human bound is tighter than the engine bound. A bot tracking the full realisation set knows exactly which menu cells are live and which its opponent's phase choices have emptied. A human sees the same information in principle — the game has perfect information — but not in practice past a handful of branches. So the bound is not only an engine tractability parameter; it is the number of realisations a person can hold. Three is probably right for humans and unnecessarily small for bots, which argues for m being a table stake rather than a rule.

Asymmetric bounds. Capping the stronger player at m while the weaker plays at $m + 1$ is a handicap with no equivalent in chess, since it changes which moves exist rather than what material is on the board. Whether it balances is empirical, and cheap to test.

11 Open questions

Open Question 1 (Is higher m actually stronger?). Under adversarial resolution a wider Q-move hands the opponent a larger menu, so width may be self-limiting: past some point it could be a liability. If the optimal m for a perfect player is 1, the ladder of §10 remains a teaching structure but stops being a ranking, and the variant's quantum content becomes decorative. This is the question on which the design most depends. A small search at $m = 2$ and $m = 3$ on a reduced board should settle it.

Open Question 2 (Are Q-moves worth playing at all?). Since the opponent chooses every measurement outcome, a superposition may simply never resolve favourably. The presumed answer is that a good Q-move creates a position in which *both* branches are acceptable — a real and recognisable skill — and that the opponent's choice is constrained by the rest of the board. But this is an empirical claim about the resulting game and should be checked before anything else is built on it. Note that Question 1 is the quantitative form of this one.

Open Question 3 (Does phase ever bite? — *answered affirmatively*). Reconvergence requires that everything other than the interfering branch agree across two histories, which on a chess position looked stringent. Appendix A exhibits a position where it holds *by construction*: an opening transposition, in which two histories play the same moves in different orders, so that the residual board is identical and there is nothing left over to serve as a which-path record. Opening theory is full of these, so interference is not rare in this game and has an identifiable habitat.

The subsidiary worry — that bare signs on integer path counts would make cancellation a coincidence rather than something a player could engineer — also resolves, with one condition.

The signs must be per Q-move, not per move. If a sign attached to a move once and for all, both orderings of a transposition would accumulate the same product and the diamond could only interfere constructively. The grammar of §6 already gives signs per branch of each Q-move, so this works as written; but the property is load-bearing and was not previously stated.

What remains open is the frequency and tactical value of such positions in real play, which is an empirical question rather than a structural one.

Open Question 4 (Interference can annihilate the support). Found by writing the trace of Appendix A rather than by reasoning about the design. In that trace a single minus sign cancels *every* surviving cell, so the support after the Q-move is empty, and an empty support is not a position. The rules have no answer. Two candidate repairs:

- a Q-move is illegal if it would empty the support — an amplitude-level analogue of Definition 5.5’s requirement that both answers be live;
- or the sign discipline is constrained so that total cancellation is unreachable.

Note that selective deletion — emptying one cell of the opponent’s menu while keeping another — requires the interfering Q-move to be entangled with whatever distinguishes the cells. In Appendix A’s trace the two diamonds are independent, so Black’s signs cannot be conditioned on White’s choice and the cancellation is necessarily total. A trace in which the diamonds interact is the natural sequel and is what Principle 8.1 needs in order to be a tactic rather than a curiosity.

Open Question 5 (Shared source or not?). Axis 5. Requiring the branches of a Q-move to share a source square gives Cantwell’s split move, which is easier to visualise and removes most bookkeeping. Permitting arbitrary sets gives the Dorbec–Mhalla reading, which is more expressive and allows branches to differ in material and tempo. We suspect the shared-source restriction is right for a playable game and wrong for the study of the underlying combinatorics.

Open Question 6 (What happens on overflow?). If a Q-move would take the support past its cap: declare the move illegal, force a measurement by the mover, or force one by the opponent. The first is cleanest; the third has an appealing symmetry with the measurement move but risks making wide Q-moves unplayable.

Open Question 7 (Is the coherent-pair condition too strong?). Condition (ii) of Definition 5.5 requires the yes-space to contain a coherent pair. It may rule out measurements a player legitimately wants, particularly late in a game when the support is small. It may be that condition (i) alone suffices.

Open Question 8 (Draws, repetition and the fifty-move rule.). Repetition detection over sets of realisations is not obviously the right notion — repetition in every realisation, in some, or of the set itself? A coherence budget, decaying with each Q-move, would sidestep the question by bounding game length directly, at the cost of another parameter.

Open Question 9 (Does the referee’s rules engine factor?). The referee must carry the full classical rules, per-realisation feasibility, the win condition, the branching bound and Definition 5.5. How much of this can be expressed as spatial formulae over the board process, so that one matcher serves the whole engine? The atom already serves three roles (§9); the question is how far that extends.

Open Question 10 (Measuring versus non-measuring interaction, in general.). Out of scope here but raised by the design: is there a polymorphic interaction primitive for interactive GSLTs, of which measuring and non-measuring receives are instances? The present game suggests the two flavours differ in their effect on state the pattern does not mention, which is a semantic rather than syntactic distinction and may be the right handle.

12 Conclusion

The design reported here is small. A player may superpose at most m moves; may instead spend a turn asking a yes-or-no question about the board, drawn from a language of one atom form and two restricted connectives; and the opponent answers. There is no chance, no hidden information, and no check.

What recommends it is that each restriction was forced rather than chosen for taste. Adversarial resolution follows from wanting the variant to answer the quantum Zermelo result rather than dodge it. Coarse observables follow from adversarial resolution, given that the board is globally entangled. The graded-identity wildcards follow from wanting coarseness to be a dial rather than a special case. The restrictions on the two connectives follow from wanting each to do one job. And the semantic legality condition follows from noticing that the property one actually cares about — that no measurement isolates a realisation — is a property of the current support and cannot be enforced by a grammar at all.

What remains genuinely unsettled is whether the game is any good, and that reduces to Question 1. If the optimal branching bound for a perfect player is one, the design is an elegant apparatus around an empty centre. If it is greater than one, then the branching bound is a capability, the ladder is a ranking, and there is a new game here.

A A worked trace: the Italian transposition diamond

A.1 Choosing an opening

We want a trace in which both players play both sides of a book branch point. Two constraints narrow the choice, and the second is the one that matters.

Both branches should be feasible in every realisation, so that the branching is uncontaminated by filtering. And the branches must be able to *reconverge*: a Q-move that merely fans out gives a tree, and a tree has no interference in it. What we want is a *diamond* — two distinct move orders arriving at the same classical position, so that two amplitudes land in the same cell. Opening theory is the natural place to look, because it is full of transpositions.

Italian Game, Giuoco Pianissimo. After 1.e4 e5 2.Nf3 Nc6 3.Bc4 both players have a book fork, and each player’s two moves commute. Black may play ...Nf6 (Two Knights move order) or ...Bc5 (Giuoco Piano move order); White may play d3 (Pianissimo) or c3 (main line). The two canonical orders

```
1.e4 e5 2.Nf3 Nc6 3.Bc4 Bc5 4.c3 Nf6 5.d3
1.e4 e5 2.Nf3 Nc6 3.Bc4 Nf6 4.d3 Bc5 5.c3
```

are the same position by transposition, and both are ordinary theory. So each player has a two-path interferometer built entirely from book moves, and the two are independent: no piece contends for a square, nothing captures, nothing checks. That independence is deliberate — it makes this the simplest trace that is not trivial, and it shows up in Appendix B as two disconnected components.

A.2 Notation

Squares are written algebraically as sugar for the (i, j) of §5: (e2) is (5, 2), file then rank. Pawns are indexed by starting file, so the e-pawn is Pawn(5). For Bishop, Knight and Rook the index L | R is queenside / kingside uniformly; for bishops this coincides with the colour-complex invariant, since the queenside and kingside bishops always stand on opposite colours.

A.3 The trace

Width bound $m = 2$ throughout.

ply 1	W	{ (e2)!(W,Pawn(5)) -> (e4) }	
ply 2	B	{ (e7)!(B,Pawn(5)) -> (e5) }	
ply 3	W	{ (g1)!(W,Knight(R)) -> (f3) }	
ply 4	B	{ (b8)!(B,Knight(L)) -> (c6) }	
ply 5	W	{ (f1)!(W,Bishop(R)) -> (c4) }	
ply 6	B	{ (g8)!(B,Knight(R)) -> (f6) , (f8)!(B,Bishop(R)) -> (c5) }	-- Black's fork
ply 7	W	{ (d2)!(W,Pawn(4)) -> (d3) , (c2)!(W,Pawn(3)) -> (c3) }	-- White's fork
ply 8	B	{ (g8)!(B,Knight(R)) -> (f6) , (f8)!(B,Bishop(R)) -> (c5) }	-- closes Black's diamond
ply 9	W	{ (d2)!(W,Pawn(4)) -> (d3) , (c2)!(W,Pawn(3)) -> (c3) }	-- closes White's diamond

Plies 8 and 9 are textually identical to plies 6 and 7: the player re-offers the same pair. What has changed is the board. In each realisation exactly one branch is now feasible, because the other piece has already moved and the source pattern no longer matches, and the Q-move does not need to know which — feasibility sorts it out per realisation. This is the source annotation of §6 earning its keep: without a pattern at the source, (g8) -> (f6) in a realisation where g8 is empty would be indeterminate rather than infeasible.

A.4 Evolution of the realisation set

Writing a realisation by the set of committed fork moves; the first five plies are classical and common to all.

after ply	support	realisations
5	1	{ }
6	2	{Nf6}, {Bc5}
7	4	{Nf6, d3}, {Nf6, c3}, {Bc5, d3}, {Bc5, c3}
8	2	{Nf6, Bc5, d3}, {Nf6, Bc5, c3}
9	1	{Nf6, Bc5, c3, d3}

At ply 8, in {Nf6, d3} the knight has moved so only Bc5 is feasible, and in {Bc5, d3} only Nf6 is — and both produce the same classical position. Likewise for the c3 pair. The support falls from four to two and each surviving cell has received *two* contributions: Black's diamond has closed. Ply 9 closes White's symmetrically.

The final position is the Giuoco Pianissimo, with the board classical. In the set semantics *nothing happened*: the game passed through a four-element superposition and came out at a

single ordinary position. Everything of interest is in the multiplicities, which the set semantics discards and the phase layer of §8 keeps. Four histories reach the final position — two orderings of Black’s pair times two of White’s — so it carries amplitude 4 with all signs positive.

A.5 Adding signs

Take Black’s two Q-moves to be

```
ply 6   B   { + (g8)!(B,Knight(R)) -> (f6) ,
              - (f8)!(B,Bishop(R)) -> (c5) }

ply 8   B   { + (g8)!(B,Knight(R)) -> (f6) ,
              + (f8)!(B,Bishop(R)) -> (c5) }
```

Tracking the two paths into {Nf6,Bc5,d3}: knight first gives $(+1)(+1) = +1$; bishop first gives $(-1)(+1) = -1$. They sum to zero and the cell leaves the support. The same computation empties {Nf6,Bc5,c3}, which is Question 4.

A.6 The measurement variant

For contrast, replace ply 8 by a measurement:

```
ply 8   B   measure (f6)!(B,Knight(R))
```

Against the four-element support this is legal by Definition 5.5: the yes-space is {Nf6,d3}, {Nf6,c3} and the no-space is {Bc5,d3}, {Bc5,c3}, so both answers are live with two realisations each, and each pair differs only in d3 versus c3, which the formula cannot see. White answers, the support halves, and Black’s diamond is destroyed before it can close.

B A ZX semantics for game traces

B.1 The question

One may reasonably ask how a variant defined over sets of realisations, which resolves measurements adversarially and never samples or normalises, maps onto a quantum computer at all.

It maps as soon as one stops trying to make the map on *positions* unitary and carries the branch register explicitly. On positions alone the step relation is not injective, because distinct histories reconverge. On positions together with a record of which branch was taken, every turn except a measurement is a permutation of basis states. The apparent non-unitarity is entirely the act of summing the branch register out when one asks for the amplitude of a position — which is not something the machine does, but something the semantics does.

B.2 Conventions

Unnormalised spiders throughout: for phase φ ,

$$\llbracket Z_\varphi^{n \rightarrow m} \rrbracket = |0\rangle^{\otimes m} \langle 0|^{\otimes n} + e^{i\varphi} |1\rangle^{\otimes m} \langle 1|^{\otimes n}, \quad \llbracket X_\varphi^{n \rightarrow m} \rrbracket = |+\rangle^{\otimes m} \langle +|^{\otimes n} + e^{i\varphi} |-\rangle^{\otimes m} \langle -|^{\otimes n},$$

so that $\llbracket Z_\varphi^{0 \rightarrow 1} \rrbracket = |0\rangle + e^{i\varphi} |1\rangle$, $\llbracket Z_0^{1 \rightarrow 0} \rrbracket = \langle 0| + \langle 1|$, and $\llbracket X_0^{1 \rightarrow 0} \rrbracket = \sqrt{2} \langle 0|$, $\llbracket X_\pi^{1 \rightarrow 0} \rrbracket = \sqrt{2} \langle 1|$.

Not normalising is not laziness: the game never samples, so a norm carries no meaning, and with this convention the amplitude of a position is exactly the signed number of histories reaching it. Green spiders are Z , red are X , and an unlabelled spider has phase 0.

B.3 Registers

Definition B.1 (Configuration register). C names every classical position reachable in the trace. In general this is Cantwell’s encoding, one qubit per square occupancy plus ancillas; for a trace whose realisations differ only by which of a set of moves has been committed, it is one qubit per commitment, which is what Appendix A uses.

Definition B.2 (Branch register). For each Q-move t of width $k_t \geq 2$, a register R_t of dimension k_t . For $k_t = 2$ a qubit; larger widths need a qudit or a binary encoding, and only $k_t = 2$ is treated in detail.

B.4 The translation

[T1] A classical turn. A width-one Q-move is a partial injection on classical positions; extended to a permutation of the basis of C it is unitary. For a move that sets a commitment bit this is a red π spider.

[T2] The split. A width-two Q-move with branches c_0, c_1 and relative sign σ compiles to: *prepare* R_t as $\llbracket Z_{\varphi_t}^{0 \rightarrow 1} \rrbracket$ with $\varphi_t = 0$ for $\sigma = +$ and π for $\sigma = -$; *commit*, by the controlled permutation $|0\rangle\langle 0| \otimes \llbracket c_0 \rrbracket + |1\rangle\langle 1| \otimes \llbracket c_1 \rrbracket$, which for commitment bits is a pattern of CNOTs from R_t (a bit flipped by exactly one branch gets a controlled flip, a bit flipped by both an unconditional one); and *filter*, projecting out the pairs (j, s) with c_j infeasible at s . The filter is the only non-unitary ingredient in the translation.

[T3] Closing a diamond. Syntactically just another Q-move, translated by [T2]. What distinguishes it is a property of the state it acts on; see Proposition B.6.

[T4] Measurement. *measure* φ with the opponent answering b compiles to the projector P_φ^b on C ; for an atom this is a computational-basis effect, a red spider of phase 0 or π per qubit. The adversarial resolution amounts to the opponent choosing *which effect is plugged in*, not which outcome is sampled. No Born rule appears anywhere in the diagram.

[T5] The trace.

Definition B.3 (Denotation).

$$\llbracket T \rrbracket = \left(\bigotimes_t \llbracket Z_0^{1 \rightarrow 0} \rrbracket \text{ on each } R_t \right) \circ \llbracket \text{turn}_N \rrbracket \circ \cdots \circ \llbracket \text{turn}_1 \rrbracket \circ |s_0\rangle,$$

and the amplitude of a final position s is $\langle s | \llbracket T \rrbracket$.

Closing a branch wire with a *green* effect is the coherent sum $\langle 0 | + \langle 1 |$, which is the game’s rule that amplitudes of histories reaching the same position add. A decohered reading would discard rather than sum, and the colour of that one effect is the entire difference.

B.5 Unitarity and coherence

Proposition B.4 (The circuit is unitary; the semantics is not). *Every turn map of a trace, on $C \otimes \bigotimes_t R_t$, is a permutation of basis states composed with the filter of [T2] — hence unitary whenever the filter is trivial — except for measurements, which are projections. The non-injectivity of the game’s step relation on positions is exactly the branch registers being summed out in [T5].*

In the vocabulary of the dilation discussion elsewhere in this programme, the branch register is the dilation, and the good case is the one in which it is *uncomputed* rather than retained, traced out or projected.

Definition B.5 (Redundant filter). Let ρ be the state before a Q-move t , with R_t correlated with C in ρ — on the support, the branch index is a function of the configuration. The filter is *redundant* at ρ if the pairs it removes already carry zero amplitude.

Proposition B.6 (Coherence criterion). *A closing Q-move acts unitarily on the support iff its filter is redundant there. In that case the composite of opening and closing leaves C independent of R_t , the branch wire disconnects, and closing it contributes the scalar $1 + e^{i\varphi_t}$. If the filter is not redundant, the branch wire stays connected to C and summing it out gives a state whose surviving amplitude depends on which branch fired — a which-path record.*

This is the formal content of the informal condition of §8, that two histories interfere only if everything other than the interfering branch agrees across them. A which-path record is, diagrammatically, a leftover edge between the branch wire's green spider and a red spider on C : Hopf does not apply, the spiders do not disconnect, and there is no scalar to be had. Erasing the record and applying Hopf are the same operation.

B.6 The trace translated

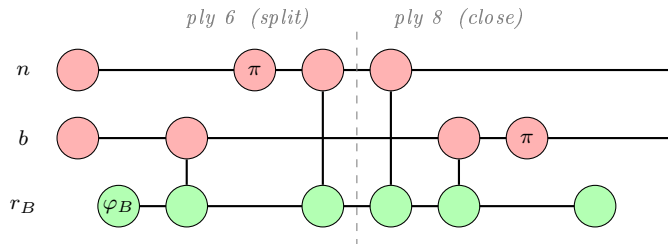
Four configuration qubits, one per commitment — n for Nf6, b for Bc5, d for d3, c for c3, all $|0\rangle$ after the five classical plies — and two branch qubits r_B, r_W . Six in total.

ply	turn	branch wire	effect on C
6	$\{ \text{Nf6}, \sigma_B \text{Bc5} \}$	prepare r_B , phase φ_B	$b \oplus = r_B; n \oplus = 1; n \oplus = r_B$
7	$\{ \text{d3}, \sigma_W \text{c3} \}$	prepare r_W , phase φ_W	$c \oplus = r_W; d \oplus = 1; d \oplus = r_W$
8	$\{ \text{Nf6}, \text{Bc5} \}$	—	$n \oplus = r_B; b \oplus = r_B; b \oplus = 1$
9	$\{ \text{d3}, \text{c3} \}$	—	$d \oplus = r_W; c \oplus = r_W; c \oplus = 1$

After ply 6 the configuration carries $n = \overline{r_B}$ and $b = r_B$: the branch register says which move was played and the board agrees. After ply 8, $n = \overline{r_B} \oplus r_B = 1$ and $b = r_B \oplus r_B \oplus 1 = 1$, both unconditionally, and r_B has left the configuration.

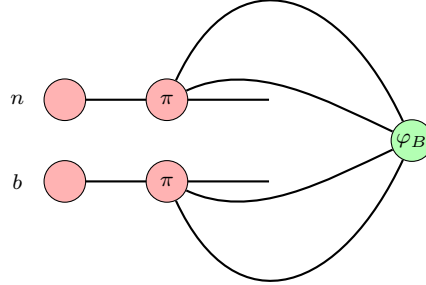
That the filter at ply 8 is redundant is immediate: in the realisation with $n = 1$ the move Nf6 is infeasible, and that realisation has $r_B = 0$, which is exactly the branch on which the controlled operation does not apply Nf6. Feasibility and the correlation agree, so the projection removes nothing and Proposition B.6 applies.

Black's diamond, drawn in full. Time runs left to right; the top two wires are n and b , the bottom is r_B .

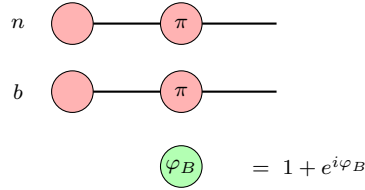


White's diamond is the same diagram on d, c, r_W with phase φ_W , and is disconnected from it.

The reduction. All green spiders sit on the r_B wire and are connected along it, so they fuse into one with phase φ_B and four legs; all red spiders on n fuse, with phases adding to π , and likewise on b :



The green spider is joined to each red spider by exactly two edges. By the Hopf law each pair disconnects, leaving



that is: n and b each become $|1\rangle$ unconditionally, and Black's entire diamond has become a legless green spider, which is the scalar $1 + e^{i\varphi_B}$.

Remark B.7 (Scalars). We compute scalars by direct evaluation rather than by chasing normalisation factors through the rewrites; the diagrams display the structure and Appendix C supplies the numbers. We note this so that no rewrite-convention is being silently relied on.

B.7 The numbers

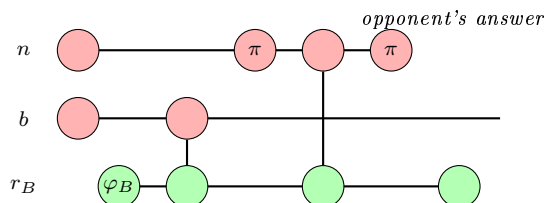
Composing the two diamonds,

$$\llbracket T \rrbracket = (1 + e^{i\varphi_B})(1 + e^{i\varphi_W}) |nbdc = 1111\rangle,$$

so the unsigned amplitude is 4 — the path count of Appendix A — and a single minus sign on either fork gives 0, which is Question 4 in diagrammatic form: the scalar spider evaluates to zero and the whole diagram with it.

B.8 The measurement variant

Replacing ply 8 by $\text{measure}(f6)!(B, \text{Knight}(R))$ substitutes a red effect on the n wire for the closing CNOTs:



The red π effect is $\sqrt{2}\langle 1|$, the projection onto *yes*. It does not fuse with the green spider on r_B , being of opposite colour and joined by a single edge; and since n is correlated with r_B , projecting n pins r_B . Closing the branch wire afterwards therefore contributes no fringe. With both signs positive, *yes* leaves $|1011\rangle$ with amplitude 2 and *no* leaves $|0111\rangle$ with amplitude 2; in both cases Black’s diamond is destroyed and the surviving factor of 2 is White’s, which still closes at ply 9.

The same six-qubit circuit, differing in one gate, gives a two-path fringe or no fringe, and the difference is whether the branch register was uncomputed or read.

B.9 The hardware reading

Strip the labels from Black’s diamond and it is a Mach–Zehnder interferometer: the branch register is prepared in $|0\rangle + |1\rangle$, acquires a relative phase, and is closed with $\langle 0| + \langle 1|$, while the configuration register is the apparatus the two arms pass through, arranged so the arms are indistinguishable at the output. In gate terms, H , then $Z(\varphi_B)$, then H , then read $|0\rangle$.

Principle B.8. A game trace compiles to one interferometer per Q-move that closes, with the two arms being two move orders and the configuration register playing the role of the apparatus. The trace interferes exactly where opening theory transposes.

For this trace: six qubits, two Hadamards, two phase gates, eight CNOTs and four X gates.

B.10 What this does and does not show

It shows the semantics is implementable and checkable, and it locates the coherence condition: Proposition B.6 turns “the histories must agree on everything else” into “the filter must be redundant”, checkable per Q-move against the current support.

It does not show that the general translation is easy. Widths above two need qudit spiders or a binary encoding, which makes the split a small circuit rather than one spider. Filters that are *not* redundant are genuine projections, and a trace containing them is post-selected rather than unitary — the same hazard identified in the graded where-clause work, reached here by a different road. And captures, promotion and castling permute the configuration register in ways commitment bits cannot express; Cantwell’s encoding handles them, at a cost in qubits this trace was chosen to avoid.

It does not settle whether the game is any good. That remains Question 1; the appendix shows only that if the answer is favourable, the resulting objects are ordinary quantum circuits of a particularly legible shape.

C Verification

`zxtrace.py` evaluates the trace twice. The *game* side implements the amplitude semantics directly: a position is a map from configurations to amplitudes, a Q-move offers each branch at each realisation, keeps the feasible pairings, and adds amplitudes on coinciding outcomes; nothing about circuits appears. The *ZX* side builds the six-qubit state vector, prepares each branch wire as $|0\rangle + e^{i\varphi}|1\rangle$, applies the controlled permutations, and closes the branch wires with $\langle 0| + \langle 1|$. The two are compared for all four sign assignments and both answers to the measurement variant.

signs	Black +1	White +1		
game :	$ 1111\rangle$:	+4	ZX :	$ 1111\rangle$: +4 agree: True
signs	Black +1	White -1		

```

game : (empty support) ZX : (empty support) agree: True
signs Black -1 White +1
game : (empty support) ZX : (empty support) agree: True
signs Black -1 White -1
game : (empty support) ZX : (empty support) agree: True

measurement variant, opponent answers YES
game : |1011> : +2 ZX : |1011> : +2 agree: True
measurement variant, opponent answers NO
game : |0111> : +2 ZX : |0111> : +2 agree: True

ALL CHECKS PASS

```

References

- [1] S. G. Akl. Quantum Chess. School of Computing, Queen’s University. <https://research.cs.queensu.ca/Parallel/QuantumChess/QuantumChess.html>
- [2] K. Burke, M. Ferland, S.-H. Teng. Quantum Combinatorial Games: Structures and Computational Complexity. arXiv:2011.03704, 2020.
- [3] K. Burke, M. Ferland, S.-H. Teng. Transverse Wave: An Impartial Color-Propagation Game Inspired by Social Influence and Quantum Nim. *Integers* 21B (A3), 2021. arXiv:2101.07237.
- [4] K. W. Burke, M. Ferland, S.-H. Teng. Quantum-Inspired Combinatorial Games: Algorithms and Complexity. *FUN 2022*, LIPIcs vol. 226, 11:1–11:20.
- [5] C. Cantwell. Quantum Chess: Developing a Mathematical Framework and Design Methodology for Creating Quantum Games. arXiv:1906.05836, 2019.
- [6] Quantum Checkers: The Development and Analysis of a Quantum Combinatorial Game. arXiv:2506.05962, 2025.
- [7] P. Dorbec, M. Mhalla. Toward Quantum Combinatorial Games. *Quantum Physics and Logic (QPL)*, EPTCS 266, 2017, p. 237. arXiv:1701.02193.
- [8] A. Goff. Quantum tic-tac-toe: A teaching metaphor for superposition in quantum mechanics. *American Journal of Physics* 74(11):962–973, 2006.
- [9] Quantum Go Machine. arXiv:2007.12186, 2020.
- [10] D. Scholten, B. Westerbaan, S. Samardjiska. Quantum combinatorial games. arXiv:2607.06550, 2026.